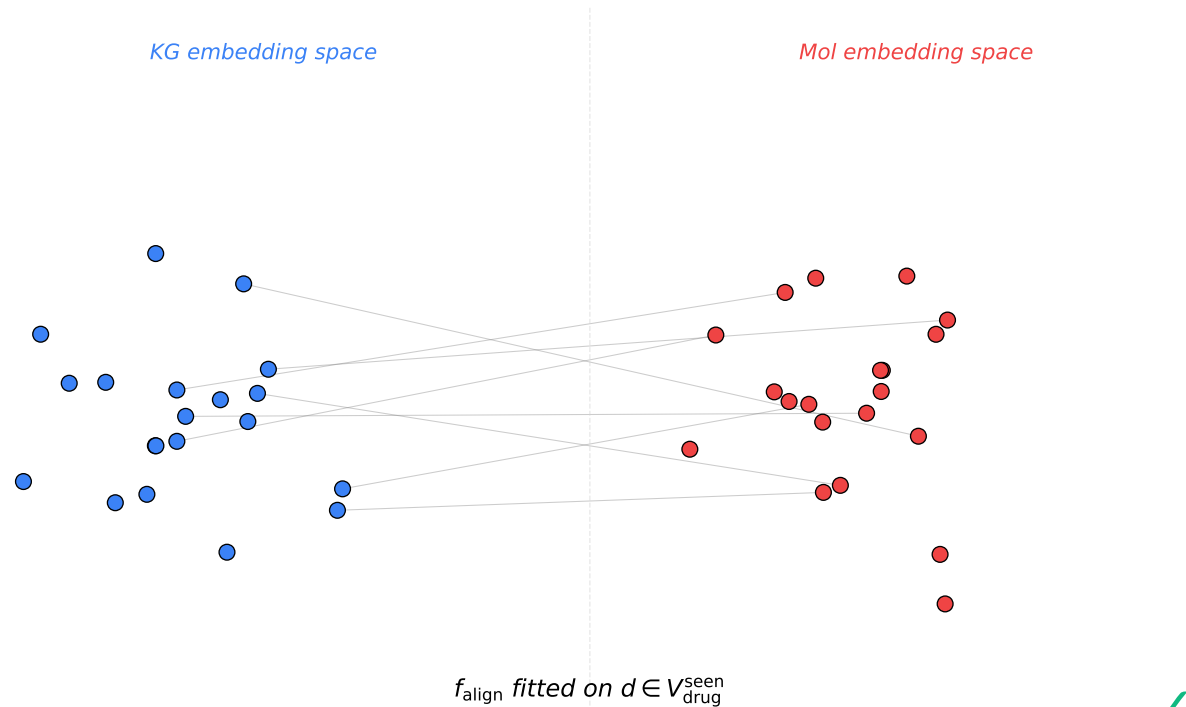
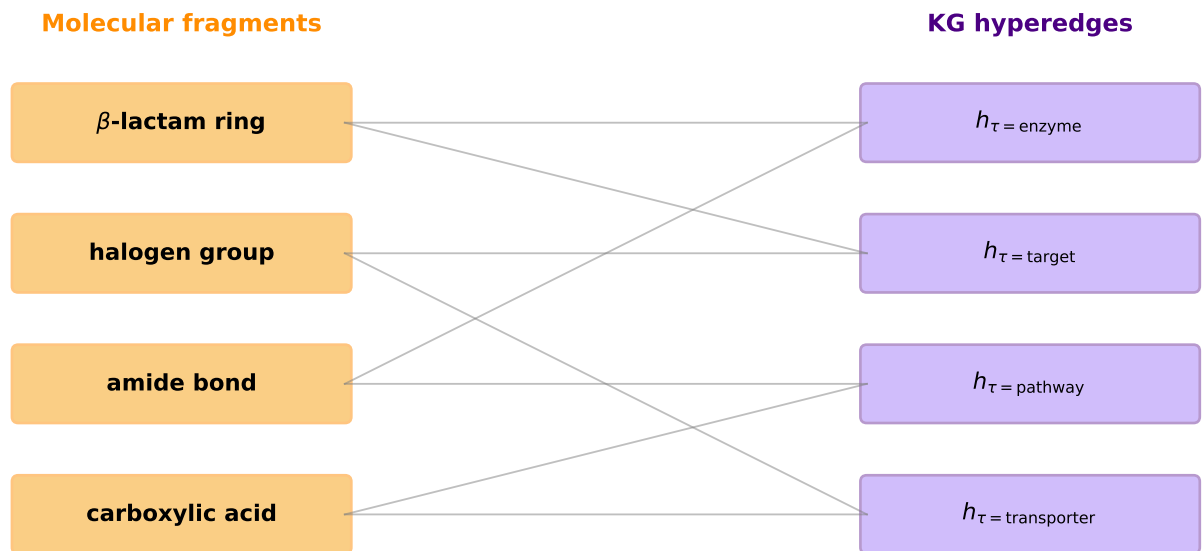


**Figure 2. Multi-modal alignment object determines cold-start transferability.**  
**Top: drug-identity alignment (MKG-FENN, TIGER) collapses on cold drug.**  
**Bottom: mechanism alignment (ours) is preserved.**

**(a) MKG-FENN/TIGER on known drugs:**  
 $e_{\text{KG}}(d) \leftrightarrow e_{\text{mol}}(d)$  aligned

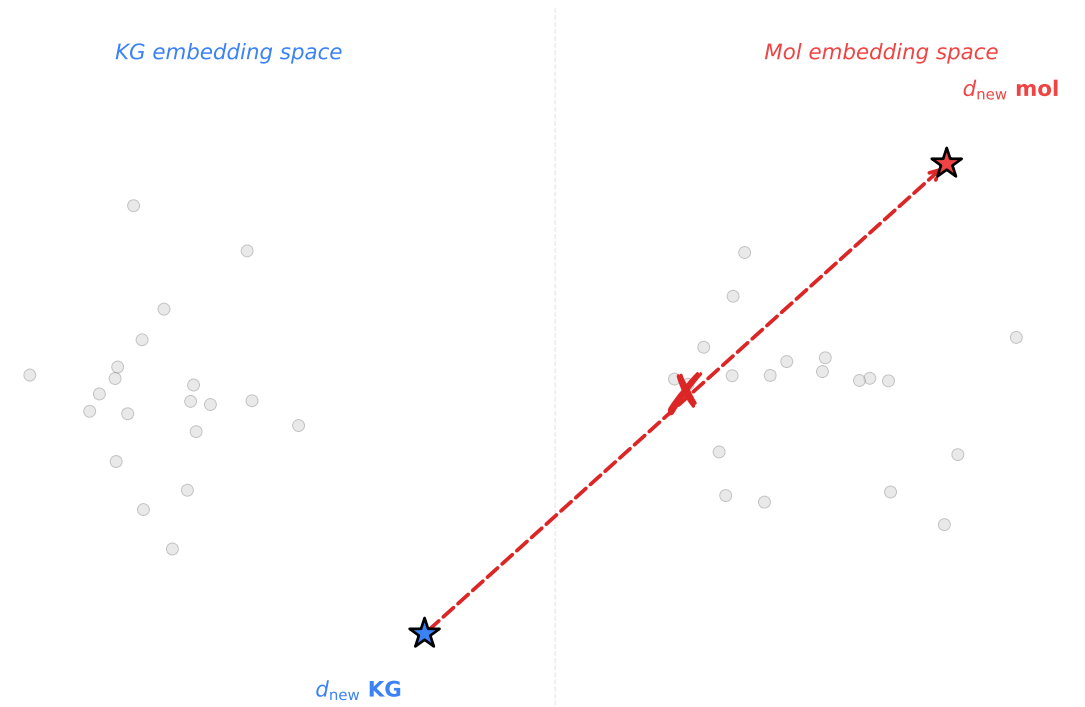


**(c) Our framework on shared units:**  
 fragment  $\leftrightarrow$  hyperedge alignment

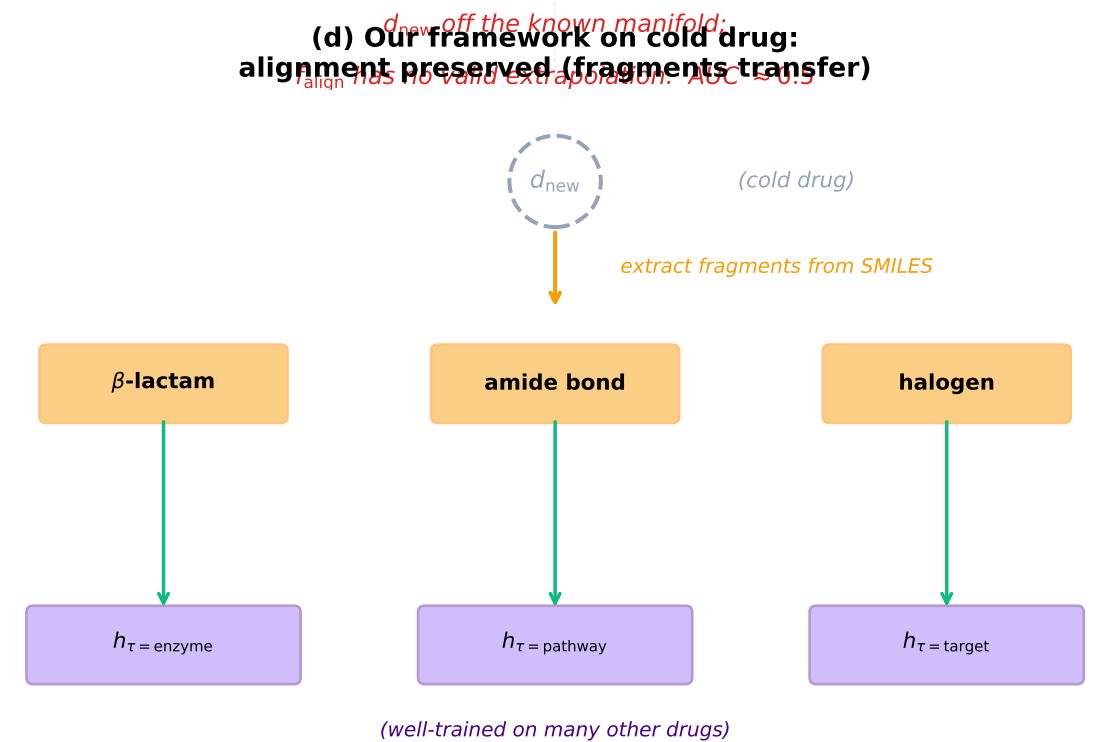


**Alignment unit = mechanism, not drug identity**

**(b) MKG-FENN/TIGER on cold drug:**  
 alignment collapses (off-manifold)



**(d) Our framework on cold drug:**  
 alignment preserved (fragments transfer)



**Fragments and hyperedges transfer beyond seen drugs**

Multi-modal DDI methods like MKG-FENN and TIGER align KG and molecular views at the drug-identity level. Under cold-start, the new drug's identity is OOD; the learned alignment manifold cannot extrapolate. Our framework aligns at the mechanism level — molecular fragments map to hyperedge mechanism units — and because fragments and mechanism units transfer beyond seen drugs, alignment survives cold-start.